

HARIRAMAKRRISHNAN RAMACHANDRAN

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PROFESSIONAL EXPERIENCE

SOFTWARE ENGINEER – HCL Technologies | Chennai, India

09/2021 – 01/2025 (3.5 Years)

- Designed and executed **DFT (Design for Testability)** strategies across mixed-signal semiconductor product portfolios, improving testability metrics by **32%** and reducing production test cost per unit by **18%** through optimized measurement methodologies and probe-card architecture definition.
- Developed **pre-tapeout verification test solutions** using **Design Simulation tools** (SPICE, Verilog simulation frameworks) to validate analog and digital electronics functionality, characterization plans, and yield analysis protocols before silicon tape-out, reducing post-silicon debug cycles by **40%**.
- Engineered innovative, reliable **multi-site Probe and Test solutions** on **high-performance mixed-signal ATE platforms**, automating test execution, Silicon functionality validation, and debug workflows in **Python** and **Visual Basic**, supporting ultra high-volume consumer portable product ramps.
- Led **Test development phase execution** including Silicon characterization, yield analysis, solution optimization, and offshore production release for next-generation capacitive sensing and optical imaging products, collaborating cross-functionally with NP development and manufacturing engineering teams.
- Drove continuous advancement of **measurement methodologies** and test automation through **32+ improvement initiatives**, establishing standardized test protocols, measurement best practices, and analytical frameworks adopted across the CSPT peer and wider Test Development community.
- Demonstrated technical leadership at project and personnel level through mentorship of junior test engineers, UAT coordination, and cross-functional stakeholder communication with hardware design, analog/digital electronics specialists, and production operations teams.

SKILLS

Analogue & Digital Electronics – mixed-signal circuit design, analog signal conditioning, digital logic verification, probe design, characterization methodology.

Test Development & DFT – DFT definition and development, Design for Testability optimization, test architecture, measurement methodology, Silicon debug and characterization.

ATE & Test Platforms – high-performance mixed-signal ATE platform programming, multi-site test execution, probe card design, test solution optimization.

Software & Automation – Python, Visual Basic, Java, C, test automation scripting, measurement data analysis, yield analysis algorithms.

Analytical & Problem-Solving – strong analytical skills, technical troubleshooting, yield analysis, statistical characterization, innovative solution design.

Design Simulation Tools – SPICE simulation, Verilog verification, pre-tapeout validation, behavioral modeling.

Teamwork & Leadership – cross-functional collaboration, inter-personal communication, project-level leadership, personnel development, stakeholder engagement.

EDUCATION AND TRAINING

**MASTER OF SCIENCE in Data Analytics – National College of
Ireland, Dublin**

02/2026

PROJECTS

Mixed-Signal Test Solution Architecture & DFT Optimization (HCL Technologies)

- Designed and implemented **DFT definition strategies** for a high-performance mixed-signal consumer sensing product portfolio, identifying and eliminating testability bottlenecks through enhanced observability nodes and controllability pathways in analog front-end and digital signal processing blocks.
- Executed **pre-tapeout verification** using **SPICE simulation tools** and **Verilog behavioral models** to validate test solution completeness, measurement accuracy, and characterization protocols, preventing **8 critical measurement gaps** that would have delayed production ramp by 6+ weeks.
- Developed multi-site **Probe and Test solution** automation in **Python** and **Visual Basic** for **high-performance mixed-signal ATE platforms**, orchestrating parallel Silicon functionality validation, yield analysis data collection, and debug workflows across **64+ test sites** simultaneously.
- Led **Silicon characterization and yield analysis** phase post-tape-out, establishing measurement protocols for capacitive sensing linearity, optical image stabilization bandwidth, and optical sensing sensitivity; optimized test parameters that improved first-pass yield by **27%**.

Test Development Continuous Improvement & Measurement Methodology Leadership (HCL Technologies)

- Championed **measurement methodology advancement initiatives** across CSPT peer group and wider Test Development community, driving **32+ standardization projects** covering probe design best practices, test-time optimization, and characterization protocols adopted company-wide.
- Designed and deployed **analytical frameworks** for yield analysis and solution optimization, leveraging **statistical analysis** and **data-driven decision-making** to identify root causes of production test failures and implement targeted improvements reducing defect escape rate by **23%**.
- Coordinated **offshore production release** activities including test solution handoff, operator training, process validation, and ongoing support; established knowledge transfer documentation and troubleshooting playbooks enabling seamless ramp to high-volume manufacturing.
- Demonstrated **technical leadership** through mentorship of junior test engineers, quarterly technical reviews with NP development teams, and cross-functional problem-solving sessions addressing analog electronics design trade-offs, digital signal processing constraints, and ATE platform limitations.

ACHIEVEMENTS & CERTIFICATIONS

- IELTS Academic: Overall Band Score **7 out of 9**, issued by IELTS Official, Test Date: 28/AUG/2023
- **Full Stack: Java Spring Boot** and Angular (Great Learning)
- Certificate of Completion in Web Development from **University of California**, Davis (Coursera)
- Certificate of Completion in Python programming from **University of Michigan** (Coursera)